UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF B.Sc. & M.Sci. WITH HONOURS

THIRD-YEAR FINAL EXAMINATION

03 21826

LH THE LIFE AND DEATH OF STARS

SEMESTER 1 EXAMINATIONS 2022/23

Time Allowed: 1 hour 30 minutes

Answer ***two*** questions. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets **[ ]**.

All symbols have their usual meanings.

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A formula sheet and two tables of physical constants and units that may be required will be found at the end of this question paper.

Answer **two** *questions. If you answer more than two questions, credit will only be given for the best two answers.*

1. Consider a red supergiant with $M = 10\,\mathsf{M}_\odot$, $R = 660\,\mathsf{R}_\odot$ and $L = 50\,000\,\mathsf{L}_\odot$.

(a) Define and derive the dynamical and thermal timescales and compute them for this red supergiant. **[6]**

(b) Consider the widely used Reimers mass loss rate,

$$\dot{M}_{\mathrm{R}} = 4 \times 10^{-13}\,\mathsf{M}_\odot\,\mathrm{yr}^{-1} \cdot \frac{LR}{M}\frac{\mathsf{M}_\odot}{\mathsf{L}_\odot\,\mathsf{R}_\odot}$$

Let us define a new "mass-loss" time scale $t_{\dot{M}} = M/\dot{M}$. Show, for the Reimer's mass loss rate, that $t_{\dot{M}}$ is proportional to the thermal timescale and compute the constant of proportionality. **[4]**

(c) Over the years, many astronomers have proposed alternative masss loss rates, calibrated to observations of red giants and supergiants. Schröder & Cuntz (2005) proposed an alternative mass-loss rate

$$\dot{M}_{\mathrm{SC}} = 8\times10^{-14}\,\mathsf{M}_\odot\,\mathrm{yr}^{-1} \cdot \frac{LR}{M}\frac{\mathsf{M}_\odot}{\mathsf{L}_\odot\,\mathsf{R}_\odot}\left(\frac{T_{\mathrm{eff}}}{4\,000\,\mathrm{K}}\right)^{3.5}\left(1 + \frac{1}{4\,300}\frac{R^2}{M}\frac{\mathsf{M}_\odot}{\mathsf{R}_\odot{}^2}\right)$$

What is the mass-loss timescale for the red supergiant described above when using Schröder & Cuntz's formula? The Sun's effective temperature is $5\,772\,\mathrm{K}$. **[6]**

(d) How do the two mass-loss time scales (Reimers and Schröder & Cuntz) compare with the dynamical and thermal time scales? Can the stellar wind be treated as being steady over thermal and dynamical times? **[4]**

(e) The adiabatic sound speed c_s is defined by $c_s = \sqrt{\Gamma_1 P/\rho}$. Convective flows usually show speeds of about $10^{-3}c_s$. Show, using appropriate approximations for P and ρ, that the time taken for a convective flow to travel from the centre to the surface is proportional to the dynamical timescale, and compute its value for the Sun and the red supergiant described above. How does this

convective timescale compare to the dynamical and thermal time scales? What does this tell us about the assumptions behind mixing-length theory? **[10]**

2. Consider a supernova progenitor with $M = 10\,\mathrm{M}_\odot$, $R = 660\,\mathrm{R}_\odot$ and $L = 50\,000\,\mathrm{L}_\odot$.

(a) Describe the Chandrasekhar limit (*not* the Schönberg–Chandraskehar limit) and give its value M_{Ch}. **[4]**

(b) Using the table of data below, compute the energy released per unit mass, denoted q, in the reaction $^{12}\mathrm{C} + {}^{12}\mathrm{C} \rightarrow {}^{20}\mathrm{Ne} + \alpha$. **[4]**

(c) The specific nuclear energy generation rate ϵ_{nuc} reaches an equilibrium with neutrino losses at $\epsilon = 10^6\,\mathrm{erg\,g^{-1}\,s^{-1}}$. What is the timescale given by q/ϵ? How does this compare with the usual nuclear timescale we defined in class, assuming a M_{Ch} core of equal parts C and O produces the star's total luminosity? **[6]**

(d) By roughly how much does the core's gravitational binding energy change by collapsing from a radius of $10\,000\,\mathrm{km}$ to $10\,\mathrm{km}$? What is the binding energy of the envelope (use $\Omega_{\mathrm{env}} = GM_{\mathrm{core}}M_{\mathrm{env}}/R_{\mathrm{core}}$) before the collapse? How does it compare to the binding energy released? **[6]**

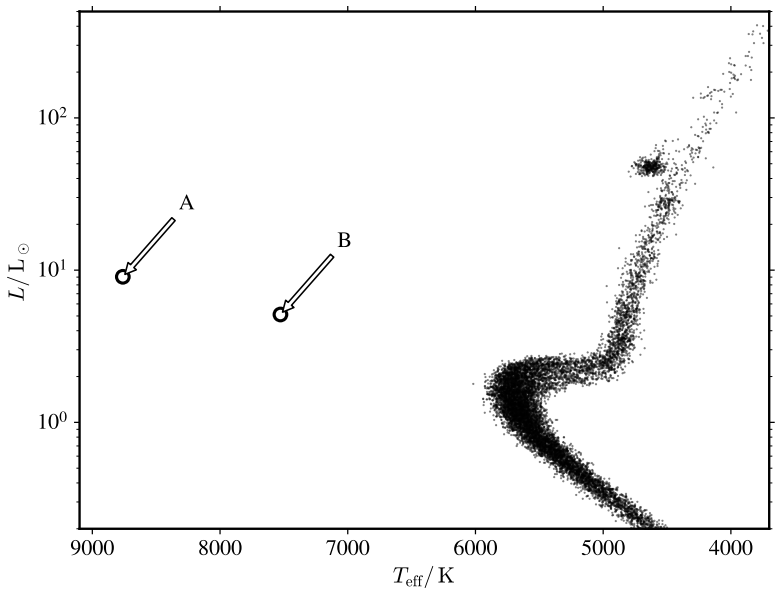
(e) One potential energy sink in driving the explosion is the photo-dissociation of metals back into α particles and protons. How much energy is required to unbind one M_{Ch} of ^{56}Fe back into 26 protons and 30 neutrons per iron nucleus? How does this compare with the gravitational binding energy of the core? **[4]**

(f) Another energy sink is the kinetic energy of the ejected envelope. Typical outflow speeds are about $10,000\,\mathrm{km\,s^{-1}}$. What would the kinetic energy of the envelope be if it were all moving outward at this speed? How does this compare with the gravitational binding energy of the core? **[4]**

(g) How is the gravitational binding energy of the core actually dissipated? **[2]**

Particle	Mass in m_u
p	1.007276466621
n	1.00866491595
α	4.00260325413
^{12}C	12 (exact)
^{20}Ne	19.9924401762
^{56}Fe	55.93493633

3. The figure shows the Hertzsprung–Russell (HR) diagram of a stellar population of initially solar composition ($X = 0.70$, $Z = 0.02$) in which two special stars are indicated as A and B.



(a) Name and describe four common features visible in the HR diagram. **[8]**

(b) Estimate the age of this population using the relation $L \propto M^3\mu^4$. **[4]**

The stars A and B have been identified as *blue stragglers* and are presumed to be the products of pairs of stars merging. i.e., two stars merged two form star A and two other stars merged to form star B. The material in the merging stars is presumed to be fully mixed throughout a merger product.

(c) Both A and B happen to be in double-lined eclipsing binaries, from which it has been determined that both stars have the same mass of $1.3\,\mathrm{M}_\odot$. If the stars both still have a solar metal abundance $Z = 0.02$, estimate their fractional mass abundances of helium Y, again using the relation $L \propto M^3\mu^4$. **[10]**

(d) Why might two merger products of the same mass have with different compositions? **[4]**

(e) Besides standing out in the HR diagram, suggest two ways in which the blue stragglers might appear different from the other stars in the population. **[4]**

Useful Formulae for The Life and Death of Stars

Equations of stellar structure

$$\frac{dm}{dr} = 4\pi r^2 \rho$$
$$\frac{dP}{dr} = -\frac{Gm}{r^2}\rho$$
$$\frac{dL}{dr} = 4\pi r^2 \rho \epsilon$$
$$\frac{dT}{dr} = -\frac{3\rho\kappa L}{16\pi a c r^2 T^3} \quad \text{(radiation)}$$
$$\frac{dT}{dr} = \frac{\Gamma_2 - 1}{\Gamma_2}\frac{T}{P}\frac{dP}{dr} \quad \text{(convection)}$$

Radiation constant

$$a = \frac{4\sigma}{c}$$

Mean molecular weight of a fully ionized mixture

$$\mu = \left(2X + \frac{3}{4}Y + \frac{1}{2}Z\right)^{-1}$$

Stefan–Boltzmann law

$$L = 4\pi R^2 \sigma T_{\text{eff}}^4$$

Astrophysical Constants and Units

Astronomical Unit	AU	1.50×10^{11} m
Parsec	pc	3.1×10^{16} m
Tropical year	y	365.242 mean solar days
Solar luminosity	$L_\odot$	3.84×10^{26} W
Absolute bolometric magnitude of the Sun	$M_{\rm bol}$	$+4^{\rm m}75$
Bolometric correction for the Sun	BC	$-0^{\rm m}08$
Apparent visual magnitude of the Sun	$m_v(\odot)$	$-26^{\rm m}74$
Solar constant	f	1.36×10^{3} W m^{-2}
Solar mass	$M_\odot$	1.989×10^{30} kg
Wien constant	b	2.898×10^{-3} m K
Hubble constant	H_o	70 km s^{-1} Mpc^{-1}
Solar radius	$R_\odot$	6.95×10^{8} m
Distance of the Sun from galactic centre		8.3 kpc
Earth mass	$M_\oplus$	5.972×10^{24} kg
Earth radius	$R_\oplus$	6371 km

Physical Constants and Units

Acceleration due to gravity	g	$9.81\,\mathrm{m\,s^{-2}}$
Gravitational constant	G	$6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
Ice point	T_{ice}	$273.15\,\mathrm{K}$
Avogadro constant	N_A	$6.022 \times 10^{23}\,\mathrm{mol^{-1}}$
		[*N.B.* 1 mole $\equiv 1$ *gram-molecule*]
Gas constant	R	$8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}} \equiv 8.62 \times 10^{-5}\,\mathrm{eV\,K^{-1}}$
Stefan constant	σ	$5.670 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$
Rydberg constant	R_∞	$1.097 \times 10^{7}\,\mathrm{m^{-1}}$
	$R_\infty hc$	$13.606\,\mathrm{eV}$
Planck constant	h	$6.626 \times 10^{-34}\,\mathrm{J\,s} \equiv 4.136 \times 10^{-15}\,\mathrm{eV\,s}$
$h/2\pi$	$\hbar$	$1.055 \times 10^{-34}\,\mathrm{J\,s} \equiv 6.582 \times 10^{-16}\,\mathrm{eV\,s}$
Speed of light *in vacuo*	c	$2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$
	$\hbar c$	$197.3\,\mathrm{MeV\,fm}$
Charge of proton	e	$1.602 \times 10^{-19}\,\mathrm{C}$
Mass of electron	m_e	$9.109 \times 10^{-31}\,\mathrm{kg}$
Rest energy of electron		$0.511\,\mathrm{MeV}$
Mass of proton	m_p	$1.673 \times 10^{-27}\,\mathrm{kg}$
Rest energy of proton		$938.3\,\mathrm{MeV}$
One atomic mass unit	u	$1.66 \times 10^{-27}\,\mathrm{kg}$
Atomic mass unit energy equivalent		$931.5\,\mathrm{MeV}$
Electric constant	ϵ_0	$8.854 \times 10^{-12}\,\mathrm{F\,m^{-1}}$
Magnetic constant	μ_0	$4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$
Bohr magneton	μ_B	$9.274 \times 10^{-24}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Nuclear magneton	μ_N	$5.051 \times 10^{-27}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	7.297×10^{-3}= 1/137.0
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12}\,\mathrm{m}$
Bohr radius	a_0	$5.2918 \times 10^{-11}\,\mathrm{m}$
angstrom	Å	$10^{-10}\,\mathrm{m}$
barn	b	$10^{-28}\,\mathrm{m^2}$
torr (mm Hg at $0\,^\circ$C)	torr	$133.32\,\mathrm{Pa}$ ($\mathrm{N\,m^{-2}}$)

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Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **must** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.